The BMW study deserves a closer look because it is the most concrete. Schuetz and colleagues took a genuine planning problem, ordering the paths a robot follows when applying sealant to a vehicle body, and tackled it with nature-inspired and hybrid quantum algorithms. The strongest results came from a classical random-key genetic algorithm, and AWS's account notes that the overhead of mapping the problem onto an annealer kept the quantum approach from reaching industrially relevant sizes. That is no mark against the study. It is the kind of honest benchmark the field needs, and a reminder never to count classical gains as quantum ones.

Two clarifications keep the picture clean. Quantum sensing, which uses quantum effects to build extremely precise sensors, is a separate field; it may matter to robots in its own right but it is not evidence that quantum processors are entering robot control. And the idea of training robot world models on quantum hardware remains speculative, without support in the work reviewed here.

Back to the night-shift robot

Return to the warehouse robot and its surprise pallet. Its wheels, joints and safety stops will keep running on fast, local classical computers, and nothing in the current evidence puts a QPU inside that loop. The plausible role is slower and humbler: a planning service that periodically hands a bounded scheduling problem (which picks, which aisles, which charging slot) to a strong classical solver and, where worth testing, to a quantum or quantum-inspired one, then checks every proposal against the robot's real limits and tries it in simulation before anything moves.

Whether that quantum branch earns its place will come down to measurable things: logical error rates that keep falling, encodings that do not drown in penalty terms, and benchmarks that beat good classical solvers on real problems. Teams can start without betting on it by framing planning problems cleanly, building honest classical baselines and keeping safety where it already works. If quantum hardware matures, those habits make it easy to plug in. If not, they will still have produced better robots.

Cost implications

The economic case for quantum-assisted Physical AI must be evaluated at the complete workflow level, not through QPU execution time or qubit count alone. Quantum processing adds an experimental planning branch; it does not eliminate classical perception, control, simulation, or safety infrastructure.

Infrastructure and access: Cloud QPU access avoids purchasing specialised hardware but introduces provider-dependent task, sampling, execution, or reservation charges. Classical compute, storage, orchestration, and simulation remain additional expenses. Owning superconducting hardware would also require cryogenics, control electronics, calibration, and specialist maintenance; this is not a practical starting point for most robotics' teams.

Formulation and integration: A substantial engineering cost arises before execution: selecting bounded subproblems, constructing QUBOs, tuning penalties, and validating discretization. Annealer embedding can multiply physical-qubit requirements, while gate-based implementations incur compilation and routing overhead. Device changes may require reformulation, making portability and reproducibility important budget considerations.

Sampling and hybrid optimization: QAOA requires repeated circuit measurements and classical parameter updates. Noise, statistical uncertainty, and error-mitigation procedures can increase execution demand. Annealing likewise requires repeated samples and post-processing. Budget estimates must therefore include unsuccessful trials, parameter sweeps, feasibility repair, and validation—not merely the final successful run.

Latency and operational value: Queueing, network transfers, and processing delays can invalidate otherwise useful plans. Measure cost per feasible, deadline-compliant solution alongside solution quality and total elapsed time. A classical fallback must remain available, so quantum expenditure is initially incremental rather than a replacement cost.

Investment and benchmarking: Begin with capped experiments on slower scheduling or offline planning workloads. Compare against tuned classical and quantum-inspired solvers under equivalent constraints. The BMW study illustrates why this matters: classical optimization delivered the strongest results. Scale investment only when repeatable improvements in throughput, energy consumption, or planning quality exceed integration and operating costs; fault-tolerant resource estimates remain workload-dependent.

Risks and mitigations

Quantum-assisted Physical AI combines uncertain optimization outputs with systems capable of physical harm. Risk controls must preserve the article's architectural boundary: quantum proposals remain advisory; actuation and safety remain local and classical.

Encoding and constraint risk: Discretization, unsuitable penalty weights, and annealer embedding can distort the original problem. A low-energy solution may still violate collision, torque, battery,